natural selection over millions of years of evolution.

2. Algorithmic calculations are not affected by the materials from which you build the calculator. Whether you build an abacus from wood, iron or plastic, two beads plus two beads equals four beads.
3. Hence there is no reason to think that organic algorithms can do things that non-organic algorithms will never be able to replicate or surpass. As long as the calculations remain valid, what does it matter whether the algorithms are manifested in carbon or silicon?

True, at present there are numerous things that organic algorithms do better than non-organic ones, and experts have repeatedly declared that something will 'for ever' remain beyond the reach of non-organic algorithms. But it turns out that 'for ever' often means no more than a decade or two. Until a short time ago, facial recognition was a favourite example of something which even babies accomplish easily but which escaped even the most powerful computers on earth. Today facial-recognition programs are able to recognise people far more efficiently and quickly than humans can. Police forces and intelligence services now use such programs to scan countless hours of video footage from surveillance cameras, tracking down suspects and criminals.

In the 1980s when people discussed the unique nature of humanity, they habitually used chess as primary proof of human superiority. They believed that computers would never beat humans at chess. On 10 February 1996, IBM's Deep Blue defeated world chess champion Garry Kasparov, laying to rest that particular claim for human pre-eminence.

Deep Blue was given a head start by its creators, who preprogrammed it not only with the basic rules of chess, but also with detailed instructions regarding chess strategies. A new generation of AI uses machine learning to do even more remarkable and elegant things. In February 2015 a program developed by Google DeepMind learned *by itself* how to play forty-nine classic Atari games. One of the developers, Dr Demis Hassabis, explained that 'the only information we gave the system